Assignment - AI Prototyping Intern

Introduction

This assignment is designed to evaluate how you use AI tools to quickly prototype, refine, and understand code. The focus is not on building production-ready software, but on showing your problem-solving process, how you collaborate with AI, and how you handle debugging when things break.

Please read the entire document thoroughly before you start your assignment.

There are three parts:

1. **A small web utility** (to test your ability to build simple user-facing tools).
2. **A small data utility** (to test your ability to process and visualise/parse data).
3. **Documentation** (to explain your workflow, AI prompts, manual changes, and how to run the code).

Prior experience or polished design is not important. What matters is that you:

- Build something that works.
- Use AI effectively.
- Understand and explain your own code.

If you are unsure what to use, you may try:

- GitHub Copilot (free with a verified student account)
- ChatGPT (free tier)
- Gemini (free tier or Pro for student accounts)

Avoid using paid services; free tools are sufficient for this assignment.

You will have 2 days to work on this assignment. You must submit the assignment within 48 hours of receiving this assignment.

At the end, you will submit your codebase, a README.md file, your AI chat history, and a short screen recording on GitHub.

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Part 1: Small Web Utility (Choose ONE)

Pick *one of the two* and build a working prototype:

- **File Upload Portal:**
  - A simple web page where a user can upload a .log, .pdf, .csv, .zip file only, enter the log date/time, description and uploader's name.
  - Store the file and metadata in any simple structure.
  - Show a confirmation page listing uploaded files.
  - Users should be able to see a list of all files on the portal and download any file already uploaded to the portal.

- **Weather Data Viewer:**
  - A simple web page that fetches current weather for a single city from at least one free API (you can choose the city and API) and displays temperature, humidity and wind speed.
  - Users must be able to change cities.
  - Show the data on a dashboard - try to make sure it looks good.
  - Plot graphs if applicable in any place (eg, probability of rain throughout the day, change in temperature throughout the day, etc).

Part 2: Small Data Utility (Choose ONE)

Pick *one of the two* and build a working script:

- **Python CSV Plotter:**
  - A Python script that loads a CSV file, lets the user choose 2 columns (for X and Y axes), and plots the data.
  - Users must be able to choose the type of graph - line chart or bar graph.

- **NMEA Sentence Decoder:**
  - C or Python program that reads two standard GPS NMEA sentences (GPGGA and GPRMC) and prints out the parsed fields.
  - Users must be able to paste a sentence that should be then formatted into a table.
  - Refer to this GPGGA webpage and this GPRMC webpage for more details.

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Part 3: Documentation

Write a brief README for whichever utilities you built. It should:

- Explain what the program does.
- List of requirements (OS, libraries, etc.) and how to run it.
- Mention which AI tools were used and what you changed manually.
- The prompts you used for the AI tools.
    - If you're using GitHub Copilot, you can download your chat session on VSCode as a JSON file. Hit *Ctrl+Shift+P* and search for *Export Chat*.
    - If you're using ChatGPT, you can download your entire chat as a PDF.

Submission Instructions

Publish the following to a **private repository** on GitHub:

- Your entire codebase.
- An .mp4 screen recording of the utilities full functionality, along with the installation procedure.
- Your AI prompts in JSON or PDF formats only.
- **Maintain a proper commit history**:
    - Don't put everything into one big commit.
    - This helps us follow your journey, not just look at the final code.
    - We don't count how many times you used AI vs. made manual changes.
    - We don't judge you on commit style or wording. Feel free to have a messy commit history.

Invite itadmin@aeronsystems.com to your repository as a collaborator.

Please send an email to hrd@aeronsystems.com once you have completed providing your relevant details. Attach a PDF version of your README file in the email.

Important

- You may be asked to present the software you created to check your understanding of the codebase.
- You will be provided with a laptop with Debian (ARM) or Windows (x86) to clone your repository and run the software in front of us.
- You will be given access to the device for 15 minutes before your presentation.
- Commits made after the given 48 hour window will result in disqualification.

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Core Assessment Criteria

Your work will be evaluated based on the following points:

1. Functionality (40%)

- **Part 1 (Web Utility)**
  - User interaction works as expected.
  - Data is stored, retrieved, and/or displayed correctly.

- **Part 2 (Data Utility)**
  - Handles user input.
  - Produces valid and readable output.

2. Use of AI Tools (20%)

- Clear evidence of AI-assisted development (e.g., prompts, generated code).
- Documentation of edits or refinements you made to the AI-generated code.
- Demonstrates an ability to prototype quickly with AI while maintaining understanding.

3. Code Quality & Structure (15%)

- Code is readable, with meaningful variable names and comments where needed.
- Proper use of file structure (e.g., separation of HTML, CSS, Python scripts, or data files).

4. Documentation (15%)

- README clearly explains:
  - What the program does.
  - How to set up and run it.
  - Dependencies (libraries, APIs, environment).
  - AI prompts used and manual changes made.
- Includes screenshots or screencasts demonstrating working utilities.

5. Presentation & Understanding (10%)

- Ability to present and explain the code and functionality during evaluation.
- Shows a clear understanding of how AI-generated code works.
- Can explain the reasoning behind manual changes and final design decisions

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Bonus Criteria (optional)

6. Ease of Installation & Portability (+5%)

- Code runs on Windows, Linux with minimal setup.
- Clear setup instructions (requirements.txt, environment.yml, or installation steps).

7. Ease of Use (+5%)

- User interface (web page or CLI) is intuitive and simple.
- Minimal steps required to perform the main task (uploading, viewing data, plotting, etc.).

8. Scalability & Extensibility (+5%)

- Code design allows easy extension (e.g., adding more APIs, handling larger files).
- No hard-coded values where configuration would be more flexible.

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